Thermal Camera

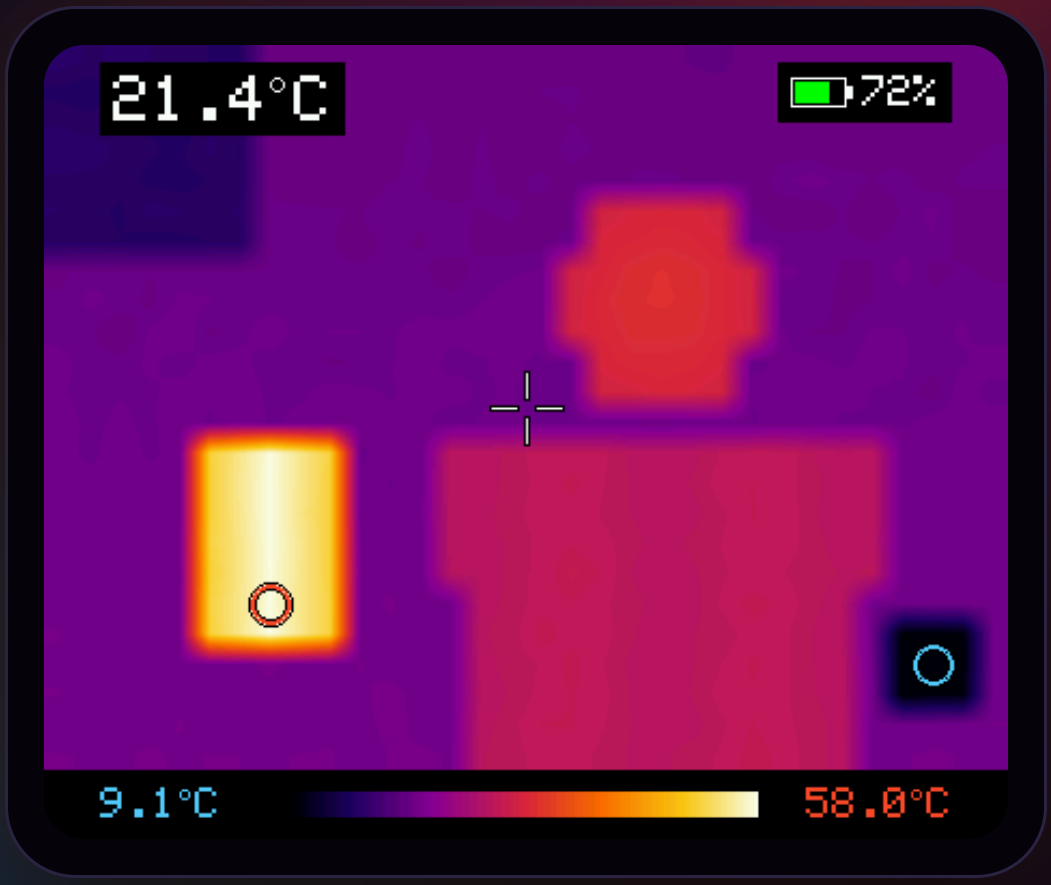
See heat in real time. Aim, read the temperature, save a picture, and watch it on a wireless second screen.

16 frames / s

32 × 24 sensor

°C / °F

Wireless screen



01 QUICK START

Point, read, save.

1 · Turn it on

Short-press PWR. The live picture appears after a moment.

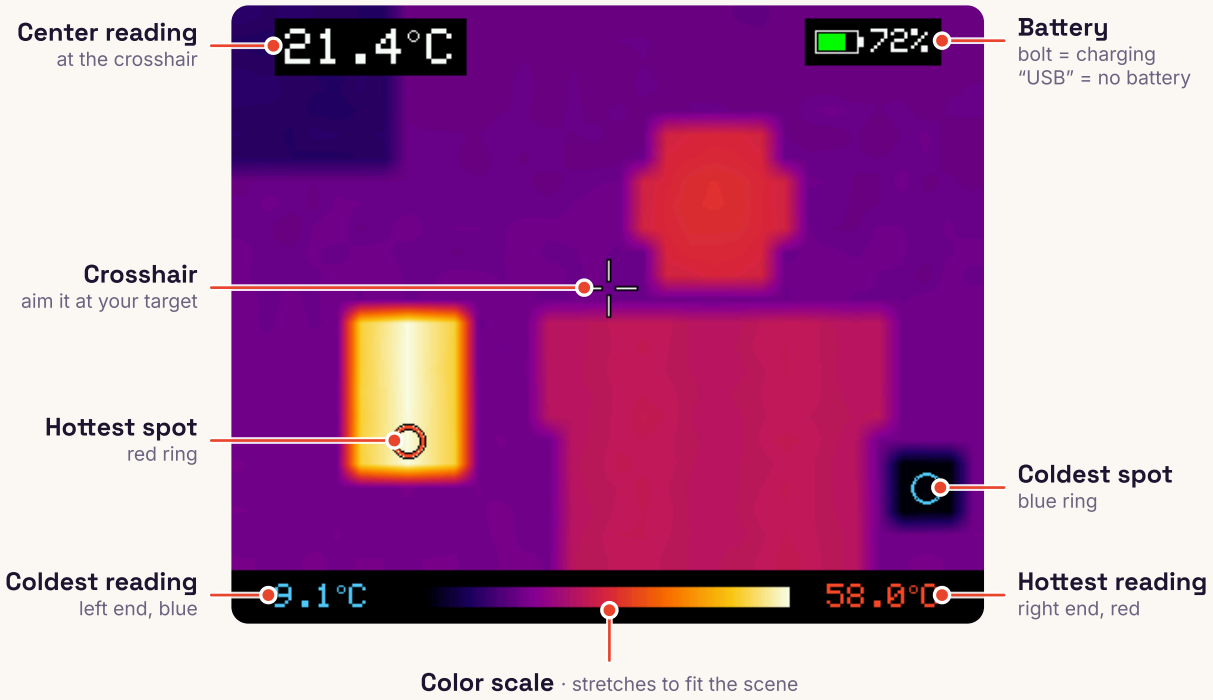
2 · Aim

Point the sensor at what you want to measure and put it under the **crosshair**.

3 · Save

Short-press PWR again to save a picture to the microSD card.

READING THE SCREEN



Rings follow the extremes. The red and blue rings sit on the hottest and coldest patch in view. They only move when another spot is clearly hotter or colder, so they don't jitter.

Smoothed, not slow. The picture refreshes 16 times a second and is gently smoothed, so readings stay steady while you aim. It still follows a moving object within about 0.1 s.

02 CONTROLS

Two buttons and a touchscreen.

BOOT changes how the picture looks, PWR turns the camera on and saves pictures, and a sideways swipe switches between live view and saved pictures.

Live camera

BOOT short press	Next color palette: Ironbow → Rainbow → Hot → White hot → Black hot
BOOT hold ~1 s	Switch between °C and °F
PWR short press	Save a picture to the microSD card. When the camera is off, this turns it on.
SWIPE → left to right	Open the picture viewer (newest picture first)

Picture viewer

TAP left half	Previous (older) picture
TAP right half	Next (newer) picture
BOOT short press	Previous (older) picture
← SWIPE or PWR	Back to the live camera

Anywhere

- **Long-press PWR** to switch the board off. The power chip does this itself.
- **Tap the screen** to keep the camera on during the auto-off countdown, or to wake a dark screen.
- A wake-up touch or press **does nothing else**: no palette change, no picture.

Remembered settings

Your palette and °C/°F choice are kept after power-off.

Swipe tips

Swipe across at least a sixth of the screen, mostly sideways. Short, straight taps count as taps.

03 PALETTES & UNITS

Choose how heat looks.

Every palette maps the coldest thing in view to the left end of the bar and the hottest to the right. Press BOOT to step through them.

Ironbow
default · all-round



Rainbow
most contrast



Hot
warm tones



White hot
classic grayscale



Black hot
inverted grayscale



Hold BOOT for about a second to switch between °C and °F. The color scale always spans at least 3 °C, so sensor noise on a plain wall doesn't look like a rainbow.

04 SAVING PICTURES

One press, two files.

Put a **FAT32 microSD card** in the slot (cards of 32 GB or less come that way). Each short press of PWR saves into a `thermal` folder:

- **`IMG_0001.bmp`**: the screen exactly as shown, with readings and color scale (448 × 368).
- **`IMG_0001.csv`**: all 32 × 24 temperatures in °C, laid out like the picture. Opens in Excel or Numbers.

What you'll see

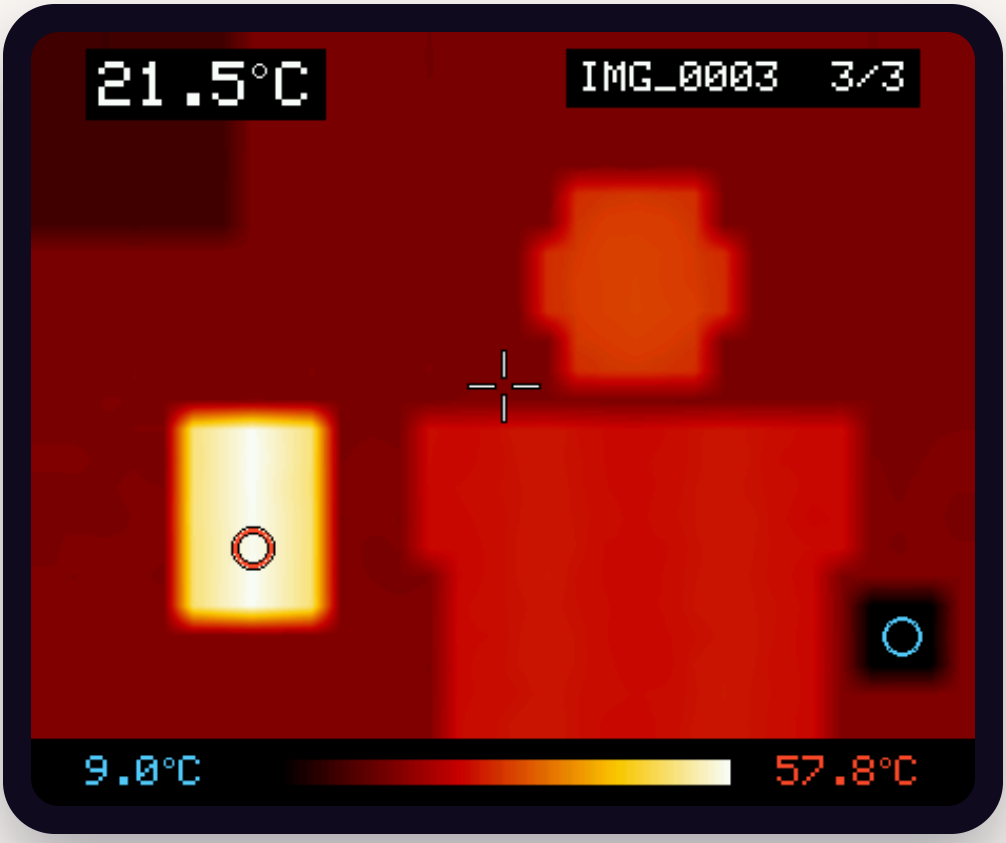
Top right briefly shows **`Saved IMG_0001`**. With no card it says **`No SD card`**.

Numbering carries on from the last picture, even after power-off.

05 PICTURE VIEWER

Look back at what you saved.

Swipe left to right on the live picture. The viewer opens on your newest picture and shows exactly what was on screen when you saved it.



The label at the top right shows the file name and its position: picture 3 of 3. Simulated screenshot.

Older

Tap the **left half** of the screen, or press BOOT.

Newer

Tap the **right half** of the screen.

Back to live

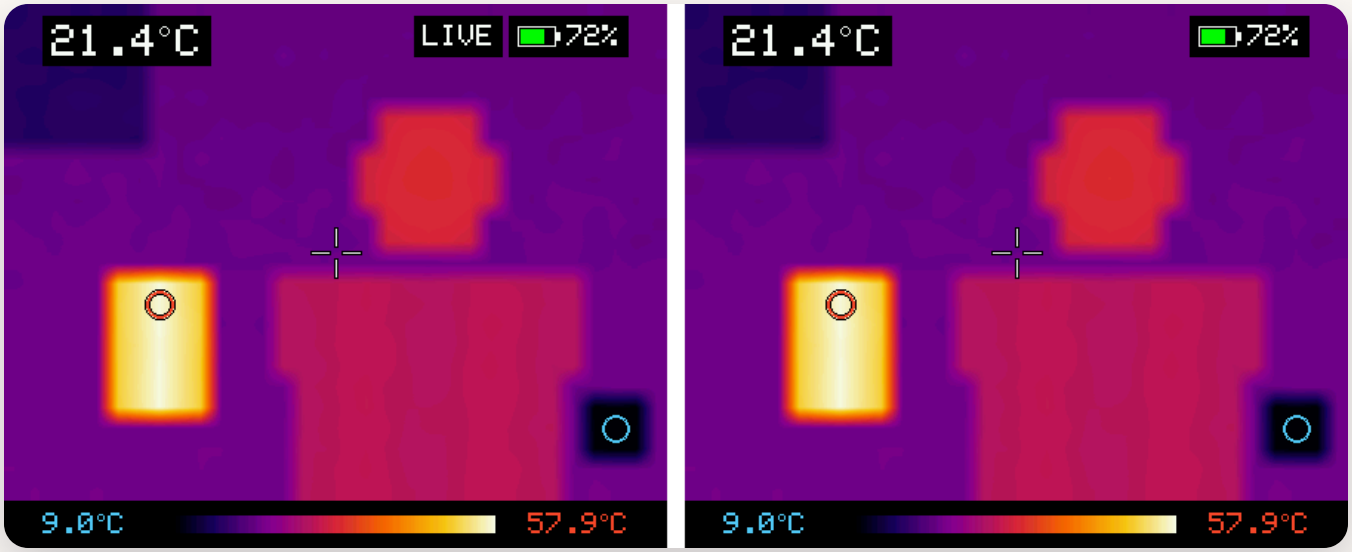
Swipe **right to left**, or press PWR.

Pictures live on the card in each board. On a wireless screen, the viewer browses the screen's own microSD card, not the camera's.

06 WIRELESS SCREEN

A second screen, no router needed.

A second ESP32-S3-Touch-AMOLED-1.8 with **no thermal sensor** becomes a live wireless screen for the camera. The boards talk directly over ESP-NOW radio; expect about a room of range.



Left: the camera's display after a tap, with the LIVE badge showing a screen is watching (it stays dark otherwise). Right: the wireless screen, showing the same picture, palette, markers and readings. Simulated.

Set it up

1. **Flash the same firmware**
   Use the same file on both boards. Nothing else to configure.
2. **Leave the sensor off the second board**
   At startup it finds no sensor and becomes a screen.
3. **Turn both on**
   The screen shows *Waiting for the thermal camera* until the camera's picture arrives.

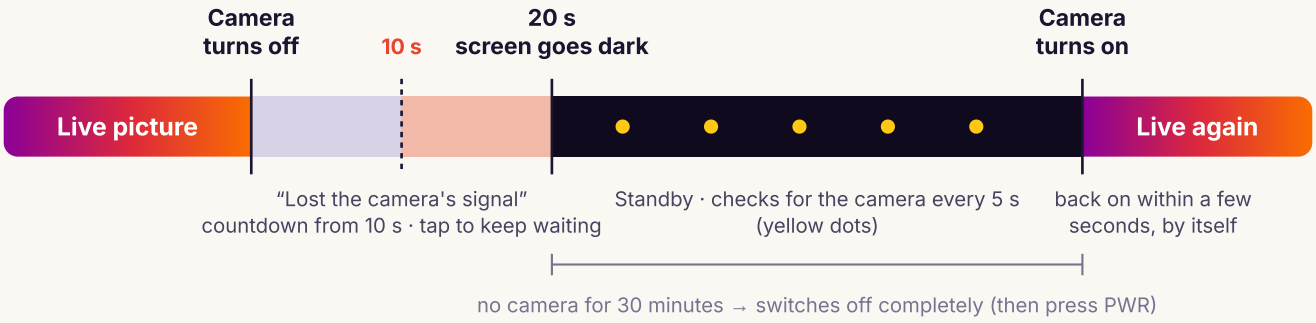
Good to know

- Palette and °C/°F follow the camera. Change them on the camera.
- PWR on the screen saves what it shows to the **screen's own** card.
- While the screen watches, the camera's own display **goes dark** to save battery. Tap it or press a button to turn it on.
- The camera only transmits while a screen is listening, and won't auto-off while one is.
- With several cameras nearby, the screen stays with the first one it hears.
- Both boards must use the same radio channel (channel 1 by default).

07 WIRELESS SCREEN · STANDBY

Turns on when the camera does.

When the camera switches off or goes out of range, the wireless screen waits a little, then goes dark. On battery it keeps listening in a low-power standby and comes back on by itself when the camera returns.



On battery

- Goes into **standby** with the screen dark, waking briefly every 5 s to ask "is the camera on?".
- **Tap the screen** or press BOOT to turn it on straight away.
- PWR works too, but is noticed at the next 5-second check.
- After **30 minutes** without the camera it powers off completely. Press PWR to turn it on.

On USB power

- Only the display goes dark; the board keeps listening.
- It lights up again **the moment** the camera's picture returns.
- Tap the screen to wake it any time.

While browsing pictures

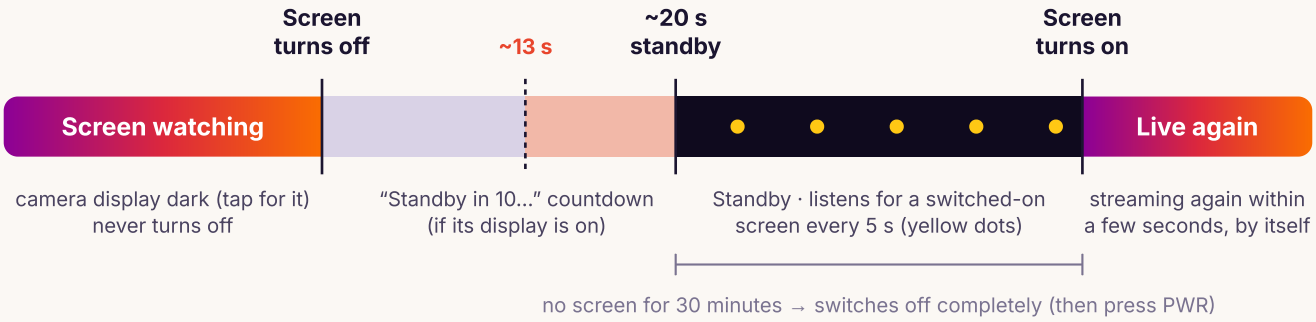
On the screen's picture viewer, the normal 60-second auto-off applies instead.

The camera's battery isn't affected. A sleeping screen's quick check gets one frame back, but doesn't count as someone watching, so the camera's own auto-off still works.

08 CAMERA · STANDBY

Sleeps when the screen leaves.

While a wireless screen shows its picture, the camera **never turns itself off**, and its own display stays dark until you tap it. When that screen is switched off or goes out of range, the camera waits about 20 seconds, goes into standby, and turns back on by itself when you switch the screen on again.



On battery

- Goes into **standby**, waking briefly every 5 s to listen for the screen.
- **Tap** or BOOT: on straight away. PWR: at the next check.
- After **30 minutes** it switches off completely.

On USB power

- The camera keeps running, with its display dark.
- It streams again **the moment** a screen connects.

Using it on its own

- Touch it or press a button after the screen leaves, and the normal **60-second** auto-off applies.
- When that runs out, it still waits in standby for the screen.

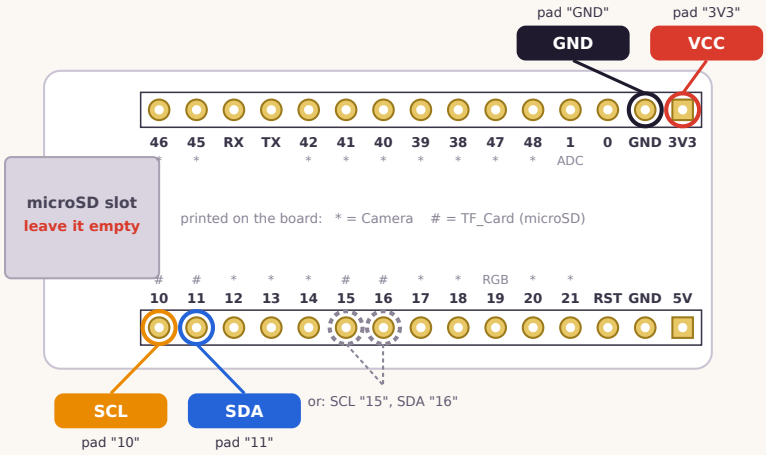
Two boards asleep? Wake either one. A sleeping camera only wakes for a screen that is switched on, so two boards in standby never keep waking each other. Tap or press a button on either board, and the other follows within a few seconds.

Standby uses some battery. The thermal sensor stays powered while the camera waits, so standby drains the battery faster than switching off. That's why it ends after 30 minutes.

09 COLOR CAMERA HEAD

Add color with a Heltec HT-HC33.

Solder the thermal sensor to a **Heltec HT-HC33** camera board and it becomes a **camera head with no screen**. The AMOLED board lays its thermal picture over its color picture.



Solder to these pads

VCC	**3V3** · the square pad
GND	**GND** · next to 3V3
SCL	**10** · next to the microSD slot
SDA	**11** · next to 10

Or use **16** for SDA and **15** for SCL. The head tries both pairs, either way round.

Leave the microSD slot empty. The pads marked **#** (TF_Card) are the slot's lines; the head uses them for the sensor.

Mount the sensor next to the lens, pointing the same way. Use 3V3, not 5V. Leave the ***** (Camera) pads, 0 and 19 alone.

FLASH BOTH BOARDS

1. **The HT-HC33**
   Program `ThermalCamHead-HT-HC33-…-0x0.bin` at `0x0` (page 14). Won't connect? Hold `USER`, press `RST`.
2. **The AMOLED screen**
   Program the current `ThermalCam-…-0x0.bin` (older screen firmware ignores color). Then turn both on.

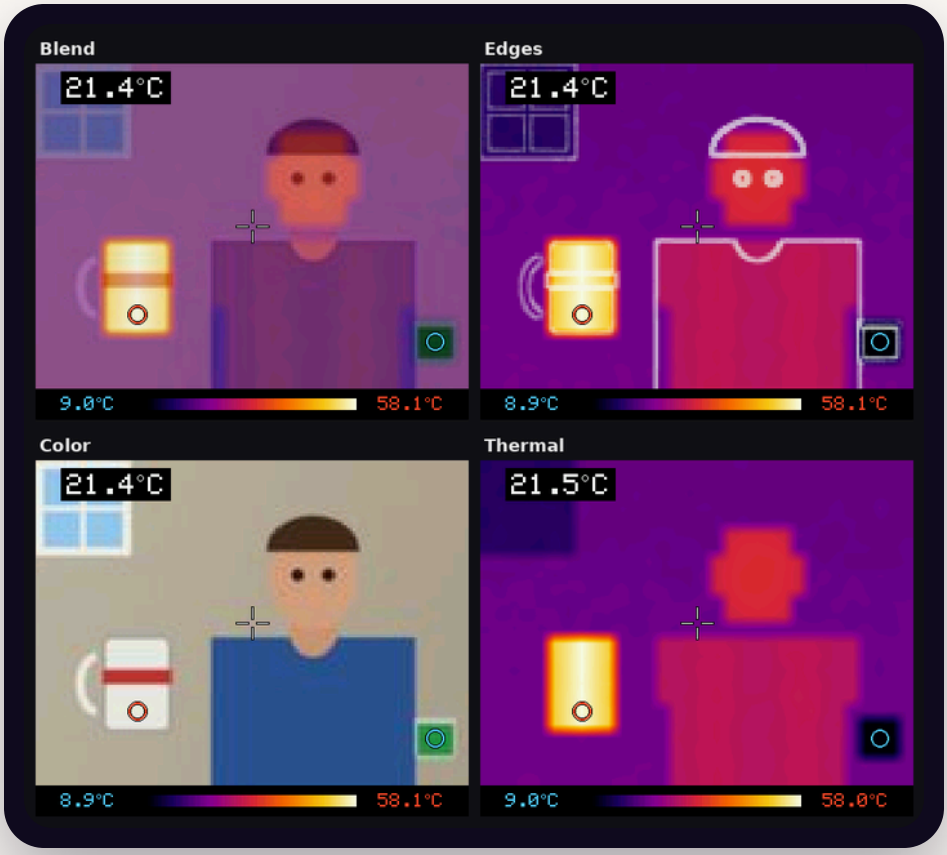
The head

- Press `USER` to switch it off; press again to turn it on.
- It only transmits while a screen is listening.
- 20 s after the screen goes away: **standby**; it comes back with the screen. Off after 30 minutes.
- Problems (no sensor, camera failed) show on the screen.

10 THERMAL OVER COLOR

Four ways to look.

Swipe **up or down** on the screen to step through the views. The screen remembers the last one.



Simulated: the screen firmware drawing the guide's made-up scene on a PC.

Blend	Thermal colors over the color picture
Edges	Thermal, with the color picture's outlines: shows *what* is hot
Color	The color picture, with the readings
Thermal	Thermal only; no color pictures are sent

With a head, BOOT on the screen sets the palette and °C/°F. PWR saves what the screen shows.

Line them up

Aim at something warm with clear edges, 1–2 m away.

- **Hold a finger still** on the picture for about a second, until *Align* appears.
- **Drag** to move the thermal picture over the color one.
- BOOT zoom in · hold to zoom out.
- PWR flip: none, mirror, upside down, both.
- **Tap** to finish: *Alignment saved.*

11 POWER & AUTO-OFF

Saves battery when you forget.

The camera counts as idle when nobody presses a button, touches the screen, or plugs in or unplugs USB. After a minute of that, it turns off.



“Tap screen to keep using”

On battery

The camera **powers off completely**. Press PWR to turn it back on. If a wireless screen was connected, it goes into **standby** instead and turns on with the screen.

On USB power

Only the **screen** turns off. Tap it, or press any button, to wake it. The waking press doesn't change the palette or save a picture.

BATTERY INDICATOR

72%

Battery level, at the top right of the picture.

Lightning bolt

The battery is charging over USB-C.

“USB”

Running from USB with no battery connected.

Someone watching remotely keeps it on. While a wireless screen is showing the camera's picture, the camera doesn't auto-off.

12 GETTING GOOD READINGS

What the numbers mean.

A thermal camera reads the heat a surface gives off. Most surfaces behave the same way, but some don't, and it helps to know which.

Accuracy

About **±1–2 °C** absolute. Readings settle after a few minutes of warm-up.

Readings assume an **emissivity of 0.95**, which is right for skin, wood, paint, plastic and food.

Shiny metal reads low

Polished metal reflects heat instead of giving it off, so it can read close to room temperature even when hot. This is normal for every thermal camera.

Fix: put a piece of matte tape on the metal and measure the tape.

Spot size

The sensor has 32 × 24 pixels. Small or distant objects cover only part of a pixel, so they read closer to the background temperature than they really are. Get closer for small targets.

Smoothing

Readings ease toward new values so they stay steady. Give the number a moment to settle after you move to a new target.

13 SAFETY & PRIVACY

- This is a **hobby thermal imager**, not a medical instrument, fire alarm or electrical-safety tester. Never use its readings as the only basis for a health or safety decision.
- The wireless link is **unencrypted and unauthenticated**. Anyone nearby with the right equipment can receive the picture (including the color camera head's color pictures), or pretend to be a camera or a screen. Don't use it for sensitive scenes.
- Saved pictures on the microSD card are **not encrypted**. Wipe the card before you lend or give the device away.
- Wire the sensor at **3.3 V** with power disconnected, and keep its wires short.

14 TROUBLESHOOTING

If something's not right.

Black screen after flashing

Unplug the board and plug it back in *without* holding BOOT. It won't run new firmware while it's still in flashing mode.

"No SD card" when saving

Use a microSD card formatted FAT32 (32 GB or less). Push it in until it clicks.

"Sensor not found"

The screen lists the I2C addresses it can see. Other devices but no `0x33`: check the sensor's VCC/GND and that SDA/SCL aren't swapped. `none`: the wires aren't where the firmware expects them.

The camera says "Wireless screen"

It didn't find its sensor at startup, so it became a screen. Check the sensor wires as above.

Wireless screen keeps waiting

Turn the camera on. Keep both boards in the same room at first, and make sure both run the same firmware and radio channel.

A board in standby won't wake

After 30 minutes in standby a board switches off completely: press PWR. If both boards are asleep, wake either one and the other follows.

"Sensor stopped" after working

A loose wire, or a long or noisy cable. Keep the sensor wires short.

Checkerboard pattern on moving objects

The shared I2C bus isn't keeping up. Lower `SENSOR_REFRESH_HZ` to 8 in `config.h` and rebuild.

Picture lags when you pan, or is jumpy

Lower `SMOOTHING` for a faster picture; raise it for a steadier one.

Swipes don't register

Swipe across at least a sixth of the screen, mostly sideways.

Text clipped in the corners

Raise `SCREEN_CORNER_RADIUS` in `config.h`.

Picture moves the wrong way

Mount the sensor pointing away from you. If it still pans backwards, change `MIRROR_IMAGE`; if upside down, set `SCREEN_ROTATION` to 3.

15 UPDATING THE FIRMWARE

Flash from your browser.

No software to install. Use the ready-made firmware file from the project's `firmware` folder, and flash the **same file** to the camera and to any wireless screen. The HT-HC33 color camera head has its own file (page 09).

1. **Open esptool-js**
   Go to `espressif.github.io/esptool-js` in Chrome or Edge.
2. **Connect**
   Plug in the board and click **Connect**. If no port appears, unplug, then hold `BOOT` while plugging it back in.
3. **Program at 0x0**
   Set the flash address to `0x0`, choose the `.bin` file, and click **Program**.
4. **Restart**
   Unplug and replug the board, or press its power button.

Flashing the full image resets stored settings.
Your palette and °C/°F choice go back to the defaults. Pictures on the microSD card are not touched.

SPECIFICATIONS

ITEM	VALUE
Sensor	MLX90640, 32 × 24 pixels, 55° or 110° view
Display	1.8″ AMOLED, 448 × 336 picture, landscape
Refresh	16 frames per second
Accuracy	about ±1–2 °C
Storage	microSD (FAT32): BMP + CSV
Wireless	ESP-NOW, about a room of range
Boards	Waveshare ESP32-S3-Touch-AMOLED-1.8, original and V2
Color head	Heltec HT-HC33 (optional)

DEFAULTS YOU CAN CHANGE

SETTING IN CONFIG.H	DEFAULT
`IDLE_OFF_SECONDS`	`60`
`SCREEN_LINK_TIMEOUT_SECONDS`	`20`
`CAMERA_DISPLAY_OFF_WITH_SCREEN`	`true`
`CAMERA_LINK_TIMEOUT_SECONDS`	`20`
`STANDBY_CHECK_SECONDS`	`5`
`STANDBY_MINUTES`	`30`
`WIRELESS_SCREEN`	`true`
`START_IN_FAHRENHEIT`	`false`
`SCREEN_BRIGHTNESS`	`200`

Changing these needs an Arduino build; see the project README.